Platypus: a complementary-learning-system language model that thinks in its sleep

Nov 3, 2026 · @Chris

Abstract

Platypus is a 171M-parameter byte-level language model that learns ten text domains in sequence with near-zero forgetting (+0.006 to +0.009 BPC) and no domain labels. It is built as a complementary learning system: a slow, protected network learns general skill, a fast entity-keyed memory holds facts, and a sleep phase between visits connects the two.

During sleep the model links facts read in different places into new conclusions, checks them against recorded evidence, revises beliefs when facts change, and dreams. A dream pairs two memories; it is kept only if each half makes the other cheaper for the model to predict, and a JEPA selector trained on dream outcomes chooses what to dream. Verified conclusions are replayed into the weights under a brake, and handed over from the fast memory once the weights hold them, with timing set by a learned model of forgetting.

On planted multi-hop facts, two-hop recall with the fast memory reaches 0.97, and REM-style recombination accepted 94 inferred facts with no probe-checked errors. When a fact changes, the fact and every chain built on it are corrected once the change is seen (1.00, old answer 0.00). Memories that fit what the network already knows consolidate about five times as readily (a schema effect). Closed-form edits with centred keys move conclusions into the weights at full size: two-hop recall from the weights alone rose to 0.55 on the GPU (0.07 without edits), and locally older memories depend less on the fast memory than recent ones, a temporal gradient (0.33 against 0.05). The transfer decayed as edits accumulated (0.28 five visits later); capping the edits' growth and making sleep length adaptive keeps most of it locally at almost no cost.

The results come from one seed, one model size and planted facts, with no matched baselines yet; we state these limits and the negative results throughout.

1 Introduction

Language models trained on one domain after another overwrite what they learned before. The usual remedies store old data, freeze parameters, or require the task to be named at test time.

Two further problems sit beside forgetting. A model told "A works with B" and separately "B lives near C" usually cannot say where A's colleague lives without reasoning in its context window: the two-hop curse. And when one fact changes, facts derived from it stay stale, which benchmarks such as MQuAKE and RippleEdits measure.

Complementary learning systems (CLS) theory offers a single account of all three. The hippocampus learns fast and sparsely, the neocortex slowly and densely, and replay during sleep moves knowledge from one to the other. Platypus implements that division in a language model and gives sleep a second job: offline inference over what the fast memory holds.

Contributions:

1. **A working CLS language model.** A 171M byte-level trunk, a fast entity-keyed memory that abstains on unknown names, label-free routing and a strict closed loop in which, after the first pass, the model learns only from replays it can verify.
2. **Inference during sleep.** Facts met in different places are chained into conclusions, grounded in recorded evidence, revised by recency when facts change, and retracted in a cascade when a premise falls.
3. **Dreams the model judges itself.** A spliced pair of memories is kept only if each half lowers the other's code length by more than random pairs do in the same sleep; a JEPA selector, trained on what the judge accepts and on measured dream outcomes, chooses among candidates.
4. **A hand-over timed by a model of forgetting.** Memories are released from the fast store when the weights answer them alone, deleted after a grace period, re-tested as ghosts and reinstated if they fade; a half-life model, calibrated out of sample, ranks what to release.
5. **Closed-form consolidation with centred keys.** Verified conclusions are written into one weight matrix in a least-squares step with the sentence frame removed from the keys, under a brake, a budget and a cap on the edits' accumulated growth.
6. **Sleep length from the model's own statistics.** NREM writes end when they converge and REM cycles end when they stop finding new conclusions, instead of following a fixed schedule.
7. **Measurements of CLS signatures in a language model:** a schema effect, ripple-consistent updates, and a temporal gradient of store dependence.

2 Related work

Continual learning. Methods divide into replay (experience replay, generative replay, dark experience replay with stored logits), regularisation (EWC and other curvature-weighted penalties, gradient projection) and parameter isolation (per-task modules and adapters).

Platypus uses all three: self-generated dream windows with stored targets, Fisher-whitened updates, and small private per-domain capacity. It differs in learning only from verified replay after the first pass.

Complementary learning systems. McClelland, McNaughton and O'Reilly (1995) and Kumaran, Hassabis and McClelland (2016) set out the fast/slow division and the role of replay. Tse et al. (2007) showed that new memories consistent with an existing schema consolidate within days. Computational CLS models exist for classification (for example DualNet and CLS-ER); we know of none that also performs inference during sleep in a language model.

Model editing and side memories. ROME and MEMIT write facts into MLP weights in closed form; AlphaEdit projects edits into the null space of preserved keys. GRACE, WISE and Larimar keep edits in a side memory. Platypus borrows MEMIT's least-squares edit, adds key centring and a cumulative budget, and feeds edits from its own verified conclusions rather than from edit requests.

Memory-augmented models. kNN-LM and Memorizing Transformers retrieve from stored activations; Titans updates a neural memory at test time by surprise. The Platypus fast memory is a delta-rule logit corrector keyed by entity, written only during sleep and silent for names it has never seen.

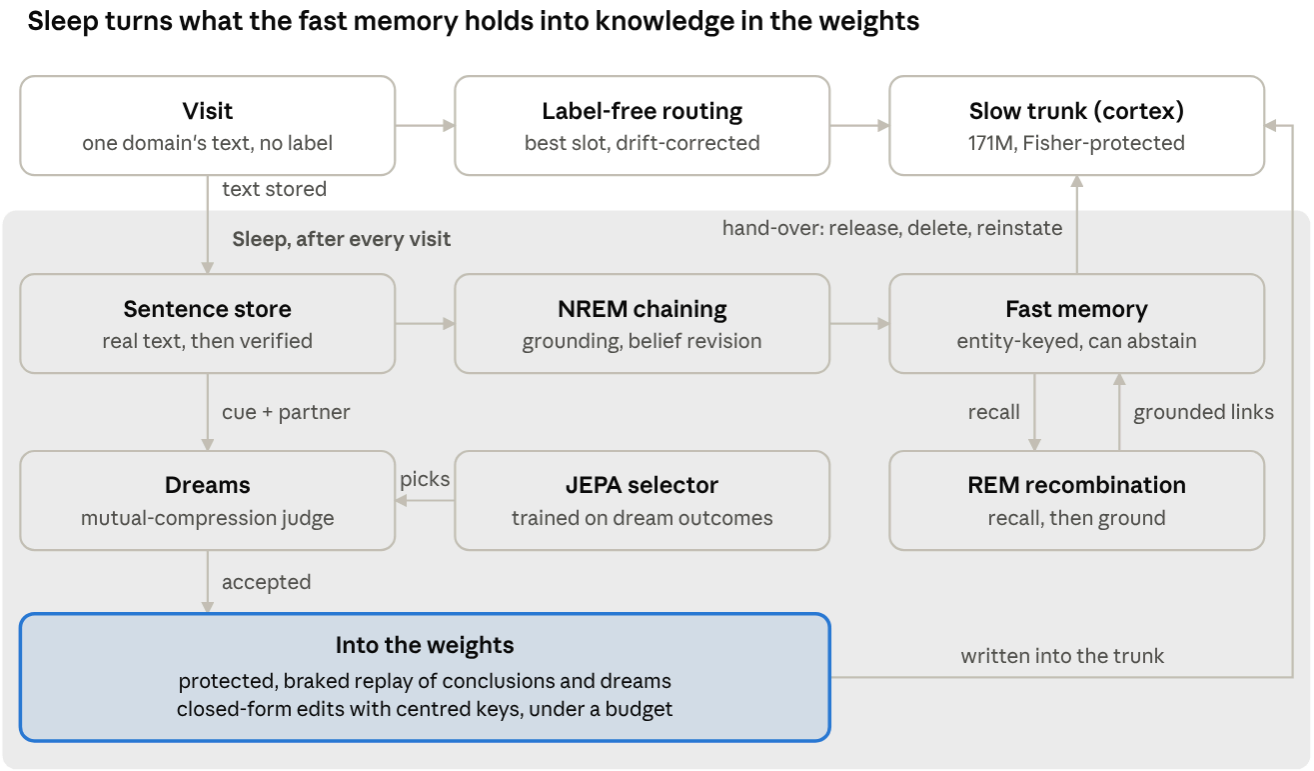
Offline computation and representation learning. Sleep-time compute lets an LLM agent reason over its context while idle; here offline inference changes a learning model's memory and weights. JEPA (Assran et al. 2023) and LeJEPA (Balestriero and LeCun 2025) learn joint embeddings without reconstruction; Platypus uses a LeJEPA objective for its dream selector. Half-life regression (Settles and Meeder 2016) models forgetting in language learners; we use it to time the hand-over to the weights. Prioritised replay by gain and need follows Mattar and Daw (2018).

Sleep for language models. The closest work is Behrouz et al. (2026), which alternates an LLM between wake and sleep phases. During sleep, knowledge seeding distils short-term, in-context memory into newly unlocked low-rank expert blocks by on-policy distillation. Synaptic pruning then resets the temporary parameters. In dreaming, the router activates experts it does not normally use to generate synthetic data, and gradient-based selection keeps the samples worth training on. The paper reports near-zero forgetting on continual translation into low-resource languages and gains on BABILong, knowledge incorporation and mathematical reasoning at much larger scale than ours. Both systems use sleep to move memory into weights and to filter dreams. Platypus differs in three ways. Its sleep draws new conclusions by chaining stored facts and audits them against their sources, so that a correction retracts what was built on the old fact. Its dreams are judged by mutual

compression and a JEPA selector rather than by gradient alignment. And the hand-over to the weights is timed by a fitted model of forgetting, with the fast memory released only when the weights are predicted to hold.

3 System

Platypus alternates visits and sleeps. A visit trains the slow trunk on one domain's text; the sleep that follows does inference, consolidation and protection.



Platypus cycle · one visit and the sleep that follows

A visit trains the slow trunk; the sleep that follows chains facts into the fast memory, recombines them, dreams, and writes what holds into the weights.

The slow trunk. A byte-level model with a 1024-byte context: byte embeddings plus hashed 2-, 3- and 4-gram tables, a causal local filter, then three gated delta-rule mixer blocks (8 heads, residual width 1088, MLP width 8,704 to 13,056). About 171M trunk parameters, plus 223M hashed n-gram parameters and one soft-capped output head per slot. Updates are Fisher-whitened against other domains' curvature, and each domain has 512 private MLP units per block and private n-gram tables.

Label-free routing. The domain is never given. Each visit's first context is scored against every slot, corrected for how far each slot has drifted since it was last trained (measured on that slot's own stored dreams). A clearly better fit routes to an existing slot; otherwise a new slot opens.

Dream rehearsal and the strict closed loop. After each visit the model writes a dream bank: self-generated windows with their top-32 predictions per byte. Later visits rehearse old slots from these banks. Real text is read only in the first pass; afterwards the model learns from replays of its memory traces that it can verify exactly against what it originally read.

The fast memory. Conclusions and recorded facts live in a sparse, delta-rule logit corrector on the trunk's pre-head features. Rows are selected by a fixed code of the recent bytes and banked by entity name, so writes are error-driven and a name the memory has never stored has no code: the memory stays silent and the model can abstain. A confidence gate (largest correction at least 3) keeps its cost on ordinary text near zero.

Cue repair. At question time, a word the fast memory does not know is mapped to the stored name it most plausibly is, before lookup. A lower-case word maps only to a name equal ignoring case, and never if it is a common word; a capitalised word maps to the nearest stored name within two edits (optimal string alignment, a swap of neighbours counting one), and only if strictly nearer than any other name. Ties abstain.

4 Sleep: inference and dreams

Each sleep runs NREM-like store writes with a REM-like recombination episode partway through, then trunk replay under a brake.

NREM chaining. A sentence store keeps real (pass 1) or verified (later passes) sentences. In the default rule-free mode, entities and relations are found from word statistics; links live in one sparse weighted table W, and every two-step path is scored at once as a sparse product S = W′W′ with log-count links and hub damping. New links are promoted by gain × need and must pay for themselves in a fixed pool. A conclusion is committed only when its premises are read again (staging).

Grounding and belief revision. A grounding audit learns which relations are single-valued. For such a relation, the most recently stated object wins once it is repeated, and conclusions built on a superseded fact are retracted, along with conclusions built on those (cascading retraction).

REM recombination. For a recorded fact "A r B", the model recalls the bridge B from memory, completes "B r′ __" for common relations and keeps the answer C only if it belongs to the bridge, judged by an appraisal margin:

```
\log p(C \mid B\, r') - \log p(C \mid \text{Someone}\, r') > m
```

with m = 2 nats. A proposal is written only if both of its links are recorded facts (grounding); otherwise it stays a hypothesis.

Dreams judged by mutual compression. A dream splices a cue sentence with a partner. It is accepted only if each half lowers the other's code length, by at least 2 bits both ways, and by more than the 97th percentile of random same-domain pairs in the same sleep. Copying is excluded, near-duplicates are dropped, and cross-domain traces that share no rare name with the cue are never dreamed.

The JEPA dream selector. An encoder head on frozen, mean-pooled trunk codes (top 3 directions removed) is trained each sleep with a LeJEPA objective: one embedding for both views, prediction to their centre, and SIGReg towards an isotropic Gaussian, weighted 0.95 and 0.05. Training pairs are verified links, splices the judge accepted, and the outcome of earlier dreams (the top third of gain changes become positives, the bottom third hard negatives). Its ranking is averaged with the judge's, and one pair per cue is dreamed.

Protected trunk replay. Conclusions and accepted dreams are replayed into the trunk at ×0.0033 of the training rate. Each Adam step is whitened by the Fisher factors of other domains; a brake rolls the sleep back if visited or old domains worsen by more than 0.03 BPC, and the dose retries lower. Replay is weighted by need (failed last test ×2, passed ×0.5) and schema fit (well predicted ×0.5, misfit ×1.5), and spaced across sleeps.

Adaptive sleep. By default (cg26n) neither phase has a fixed length. NREM writes run in chunks until the bytes of the keyed conclusions stop falling (write convergence). REM then repeats: each cycle recombines what the store now holds, and sleep ends when a cycle accepts no new conclusion (closure) or when the geometric prediction of the next cycle's yield, $a_k^2 / a_{(k-1)}$, falls below one conclusion, with at most six cycles. Locally this used 237 NREM write steps over a run against 600 on the fixed schedule and accepted 36 REM conclusions against 25, none wrong, with forgetting and BPC unchanged.

5 Consolidation and the hand-over to the weights

Release, deletion, ghosts, reinstatement. Each sleep tests tracked conclusions with the fast memory switched off. One the weights answer alone, by a 1-nat margin, at consecutive sleeps (three, or two for a schema fit) is released: erased from the fast memory and kept in trunk replay. Two passing sleeps after release, its record is deleted and only a ghost (names and domain) remains; ghosts are re-tested each sleep, and one that fails twice running is reinstated unless a source sentence has since been superseded. A released memory that fails twice is restored.

A model of forgetting. Store-off recall d visits after the last kept replay is modelled as

$$p(\text{recall}) = 2^{-d/h}, \qquad \log_2 h = \theta^\top x$$

where x holds the log of kept replays, their spacing, the bits the trunk needs for the conclusion, the previous margin and the subject's frequency. The model is fitted online and may time release only after it beats a constant rate out of sample at two sleeps running. Because fitted half-lives are short (0.2 to 0.5 visits), release ranks by predicted next-visit recall: a passing conclusion is released if it is in the top quarter of the pool, at least twice the median and at least 0.2. The fixed rule still releases beside it. Replay weights follow predicted decay relative to the pool, clipped to ×0.5–×2.

Closed-form edits with centred keys. Up to 32 verified conclusions per sleep are written into the last block's MLP output in one least-squares step, after MEMIT:

```
\Delta W = R^\top \tilde K\,(\tilde K^\top \tilde K + \lambda n\, P C_0 P +
\epsilon I)^{-1} P, \qquad \tilde K = K P
```

K holds the MLP inputs at the answer positions, R the output changes that fit each answer (found by 25 gradient steps on a free offset), C0 the key covariance of real text, and P removes the frame direction: the top singular direction of other conclusions' answer keys, about half of the edit keys' energy. An edit is applied at ×0.5 (retried at ×0.25, ×0.125), kept only if visited and old domains rise by at most 0.015 BPC, and no further edits are made once kept edits have cost 0.05 BPC on old domains.

Growth cap and keys kept back. On the GPU the uncapped edits accumulated until the edited matrix was 2.75 times its first size, and recall from the weights fell (Section 7.7). cg26n keeps the accumulated edits within 1.5 times the norm of the unedited matrix, and holds back at least 16 live conclusions from each edit to estimate the frame direction from; no edit is made below 8 + 16 live conclusions. The first two cg26m edits, made with too few conclusions to centre on, would have cost +5.9 and +7.9 BPC and were rolled back by the brake.

6 Experimental setup

Data. Ten domains of the Pile, streamed from `monology/pile-uncopyrighted` by its domain field: Gutenberg (PG-19), Wikipedia (en), ArXiv, PubMed Abstracts, FreeLaw, Github, StackExchange, USPTO Backgrounds, DM Mathematics and EuroParl. Each pass reads fresh text (33 MB per domain per pass); train and validation are split on the raw stream, so no window repeats. An early version tiled small files into both splits and reported impossibly low BPC; that leak motivated this design.

Schedule. Three passes over the ten domains (30 visits of 400 steps), the visit order shuffled after pass 1. Real text is read only in pass 1; passes 2 and 3 are closed-loop.

Probes. 150 templated chains about invented people (47 three-hop) are planted 30 times per pass into their domains' text, with 30% of two-hop chains changing in pass 2 and a few single false contradictions. A probe asks "So A is connected to" and scores the true end against 19 candidates (chance 0.05). We report recall from the weights alone (fast memory off) and with the fast memory; updated facts are scored only after their domain has been revisited. Re-DocRED documents are planted as a real-text benchmark.

Metrics. Bits per character (BPC) on held-out text of each domain; forgetting = a domain's final BPC minus its BPC right after its own last visit, averaged; two-hop recall by conclusion age (temporal gradient); held-out JEPA retrieval converted to d′ to correct for the size of the held-out set.

Harnesses. GPU runs use the full model on one 95 GB device, seed 1; a run takes four to five hours. A local CPU harness (five domains, two passes, a 2-block model of width 512, 128-byte windows, 40 probes) tests options before they reach the GPU; its seed noise is about 0.017 BPC.

7 Results

7.1 Forgetting without labels (GPU)

Across five builds, pass-1 forgetting stays below 0.01 BPC with every visit routed correctly.

Build	Pass-1 mean BPC	Pass-1 forgetting	Routing
cg26h	1.735	+0.008	10 of 10
cg26i	1.733	+0.005	10 of 10
cg26k	1.739	+0.006	14 of 14 (to visit 14)
cg26l	1.733	+0.009	14 of 14 (to visit 14)
cg26m	1.733	−0.002	13 of 13 (to visit 15)

Early domains improve after later training (cg26l: Gutenberg −0.064, Wikipedia −0.030, ArXiv −0.022), so part of the sequence shows backward transfer. With closed-form edits (cg26m) the domain average after pass 1 was 1.735, against 1.724 for cg26l.

7.2 Inference across episodes (GPU)

With the fast memory, two-hop recall rises over pass 1 as facts accumulate (0.03 at visit 1, 0.93–0.97 by visits 10–14) and three-hop reaches 0.57. REM recombination accepted 94 grounded conclusions in cg26l visits 5–10, none wrong on the probes, and supplies a growing share of conclusions (18 to 28 per sleep by visit 9). Without weight edits (cg26l), two-hop from the weights alone stays at 0.03–0.07; Section 7.7 gives the edited case.

7.3 Changed facts (GPU)

In a run that completed pass 2 (cg26e), each changed fact was corrected once its domain was revisited:

With the fast memory	Visit 10	Visit 14	Visit 18	Visit 20
Updated one-hop	0.00	0.35	0.70	1.00
Old answer preferred	0.95	0.65	0.30	0.00
Updated two-hop (derived chains)	0.00	0.10	0.40	1.00

The rise tracks the revisit order, not a slow update: locally, every changed fact is right at the first evaluation after its revisit. From the weights alone, updated one-hop reaches 0.55, level with ordinary facts (0.57).

7.4 A schema effect (GPU)

Conclusions the trunk already predicts well (the lower half by bits) pass the store-off test 0.12–0.24 of the time, misfits 0.03–0.05, at every sleep from visit 3; every one of 25 releases in cg26l is such a fit. Replay deliberately gives fits fewer replays (×0.5), so the effect is not a replay artefact.

7.5 The JEPA selector

On the GPU, held-out separation rose with training pairs (d′ 1.25 to 1.84 as pairs grew from 114 to 391, cg26k), and outcome training raised the share of dreamed pairs that later gained at least 1 bit from 0.06 to 0.33. A linear-head analysis shows the trained head matches its closed-form optimum, so the limit is data: about 390 pairs for an effective input of about 128 dimensions. Locally, 2.7 times the pairs doubled separation (d′ 1.79 against 0.84 at the same sleep). TF-IDF on names still chooses better among name-sharing sentences (0.73–0.90 against 0.44–0.68).

7.6 The hand-over (GPU, cg26l)

The model of forgetting calibrated at visit 13. Relative predicted-recall release then marked 13 and 12 conclusions (bar 0.58–0.60), where the absolute rule had marked none, and 15 conclusions were released by visit 14 against 6 in cg26k, with old-domain BPC unchanged (1.655 against 1.653). Released conclusions answered store-off half the time on re-test, close to the 0.58 predicted. Both deleted conclusions failed store-off at visit 13, as in cg26k; one failed again and was reinstated, the other recovered.

7.7 Closed-form edits (local and GPU)

Locally, centred edits gave the first durable transfer into the weights:

End of run	With centred edits	Without edits	Uncentred edits
Two-hop, weights alone	0.31	0.00	0.26
Three-hop, weights alone	0.11	0.00	0.11
Older vs recent conclusions, weights alone	0.33 vs 0.05	—	—
Edited 4+ visits earlier, weights alone	0.29	—	—
Mean forgetting / final BPC	−0.002 / 2.104	−0.012 / 2.073	+0.013 / 2.119

The uncentred column is an earlier run with stronger settings (×1, brake 0.03). Older conclusions depend less on the fast memory than recent ones, the temporal gradient CLS theory predicts once consolidation works. The cost is slower learning on later visits as the edited matrix grows about sixfold over six edits; at this width the edits' marginal gain also shrinks.

On the GPU (cg26m, seed 1), two-hop recall from the weights alone rose through pass 1, then fell in pass 2 as the edits accumulated:

Visit	5	7	9	10	11	15
Two-hop, weights alone	0.08	0.25	0.48	0.55	0.45	0.28
Three-hop, weights alone	0.06	0.06	0.19	0.34	0.30	0.17
Size of the edited matrix	41	54	61	64	76	113

Without edits (cg26l) two-hop from the weights stayed at 0.03–0.07. Each of the 11 kept edits cost old domains at most +0.0003 BPC (+0.0026 in all), and conclusions edited four or more visits earlier still answered 0.50–0.79 from the weights, so the decline is not a collapse. It coincides with the edited matrix growing past twice its first size, and released memories began to fail again: 24 of 65 had been restored by visit 15. One-hop from the weights dipped to 0.21 and recovered to 0.38, level with a run without edits.

7.8 Cue repair (local and GPU)

Two-hop with the fast memory and a damaged subject name rose from 0.00 to 0.94 (last letter dropped; one tie abstained), 1.00 (inner letters swapped) and 1.00 (lower-cased). No repair named the wrong subject in any visit, and forgetting and BPC were unchanged (−0.018 / 2.074).

On the GPU (cg26m) the same readout gave 0.79–1.00 with the last letter dropped and 1.00 with inner letters swapped or the name lower-cased, with no wrong repair in any visit.

7.9 Capped edits and adaptive sleep (local, cg26n)

The full cg26n build against the variants it combines, at the end of the local run:

End of run	cg26n	Growth cap only	Uncapped edits	No edits
Two-hop from the weights alone (typed wording)	0.26 (0.37)	0.26 (0.17)	0.31 (0.26)	0.00 (0.03)
Three-hop from the weights alone	0.22	0.17	0.11	0.00
Older vs recent, weights alone	0.27 vs 0.21	0.27 vs 0.11	0.33 vs 0.05	—
Size of the edited matrix (unedited 6.4)	14.0	13.8	39.1	—
Asked by description, with the store	0.88	—	—	0.62
REM conclusions accepted (none wrong)	34	27	24	25
Mean forgetting / final BPC	−0.009 / 2.081	−0.008 / 2.079	−0.002 / 2.104	−0.012 / 2.073
Run time	2975 s	3488 s	3606 s	3528 s

The growth cap keeps most of the transfer at almost none of the BPC cost. The same build with the fixed sleep schedule gave −0.007 / 2.075, three-hop 0.17, description recall 0.69, 28 REM conclusions and a run time of 3278 s, so adaptive sleep accounts for the gains in three-hop recall, description recall and REM yield. With edits on, conclusions made by REM passed the test without the fast memory more often than those made by NREM chaining (0.23 against 0.14 of tests; 6 against 4 released). Whether the cap stops the pass-2 decline at full size is the next GPU test.

8 Negative results

Several mechanisms that are principled on paper did not help; we report them because they bound what the design needs.

Mechanism	Result	Reading
Principal-direction input for the JEPA	final BPC +0.027 (2.101 against 2.074)	Harmful: worse dream choice damaged one domain's later learning
Platonic dream selection (dreams that move linked codes towards a shared representation weighted ×4)	alignment edge faded over 4 sleeps; −0.013 / 2.081	No effect at the leverage dreams have
LLM-JEPA term in replay	−0.017 / 2.079	Inert
REM self-exam (dream what the cortex answers wrongly)	−0.014 / 2.078	Neutral
Same-document negatives for the JEPA	−0.012 / 2.082	Neutral
Absolute replay by predicted forgetting	uniform weights	A no-op at 0.3-visit half-lives; fixed by ranking relative to the pool
Overlapping (completion) link memory	at chance on the GPU for damaged names	Completion does not scale as built; cue repair replaces it for spelling damage
Decay of accumulated closed-form edits (a share taken back each sleep)	−0.009 / 2.115; older vs recent 0.20 vs 0.10	Worse than capping the growth (2.079)

The pattern is consistent: mechanisms that change what the model learns from (more JEPA pairs, which memories are written into the weights) moved their targets, while added machinery on top of an unchanged learning signal did not. The REM analogy dreams could not be tested locally: two links do not fit the harness's 128-byte rows, so all local analogy runs were readout-only.

9 Limitations and threats to validity

- **No matched baselines.** Forgetting has not been compared, at equal compute, with plain fine-tuning, replay with the same buffer, EWC or per-domain adapters, nor inference with a retrieval-augmented baseline. The low forgetting may partly reflect these domains and this size.
- **One seed, one size.** GPU results are from seed 1 at 171M. Local seed noise (about 0.017 BPC) is as large as several effects reported, which is why the local results are read through targeted readouts.
- **Synthetic facts.** Inference, updates and cue repair are measured on templated facts about invented people. On real prose, facts stated only once are not chained, and pronouns and descriptions are not resolved.
- **The fast memory does most of the work.** Two-hop is 0.97 with it. From the weights alone it reached 0.55 on the GPU with closed-form edits but fell to 0.28 by visit 15 as the edits accumulated; the growth cap that holds it locally is untested at full size.
- **Not data-free.** The system keeps a bounded store of real sentences and its own dream windows.
- **Given boundaries.** Visits and sleeps happen at known boundaries; streaming without boundaries is not addressed.
- **Sleep compute.** Each sleep adds about 2.5 to 3 minutes on the GPU, and dream rehearsal makes a visit about 1.3 times as costly as an unprotected one; a fair comparison must charge for both.
- **Complexity.** The research code carries many interacting options; several bugs found during this work silently invalidated tests until caught (a tiling leak, a readout that misread updates, analogy rows dropped at replay).

10 Conclusion and future work

A complementary-learning-system design gives a small language model near-zero forgetting without labels, precise inference across episodes, updates that ripple to derived facts, and, with centred closed-form edits, transfer into the weights at full size

(two-hop 0.55 from the weights alone) and a first temporal gradient of consolidation. Its central open problem is the one CLS theory places at the centre: making that transfer last as it accumulates, at a cost the rest of learning can bear.

Next steps, in order:

1. **Capped edits through pass 2** (cg26n, on by default): whether the growth cap holds weights-only recall through passes 2 and 3 on the GPU (cg26m fell from 0.55 to 0.28); if not, refreshing edited conclusions instead of overwriting them.
2. **Controls and baselines:** JEPA dreams against random dreams on the same build; matched continual-learning and retrieval baselines at equal compute; three seeds.
3. **Public benchmarks:** MQuAKE for multi-step edits, a multi-hop question set, and the precision of conclusions drawn from real text.
4. **Sleep structure from statistics:** adaptive NREM and REM lengths are in cg26n; next, derive the cycle cap and the balance of NREM and REM from the same yield model, and test both on the GPU.
5. **Beyond spelling:** completion from descriptions (0.88 locally in cg26n, a readout on templated facts) and from partial context, and fact hand-over as well as conclusions.

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Compiled from memory, not re-checked against the sources: verify titles, venues and years before publication.

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